

Algebra: Function Operations & Composite Functions

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DIRECTIONS

Given $f(x)$ and $g(x)$, perform the operation or evaluate the composite function.

1 Given $f(x)=2x+3$, $g(x)=x-1$. Find:

$$(f + g)(3)$$

Answer: _____

2 Given $f(x)=x^2+1$, $g(x)=3x$. Find:

$$(f \circ g)(2)$$

Answer: _____

3 Given $f(x)=x+4$, $g(x)=x^2-2$. Find:

$$(g \circ f)(1)$$

Answer: _____

4 Given $f(x)=2x$, $g(x)=x+5$. Find:

$$(f + g)(4)$$

Answer: _____

5 Given $f(x)=x-3$, $g(x)=x^2$. Find:

$$(f - g)(2)$$

Answer: _____

6 Given $f(x)=x-3$, $g(x)=x^2$. Find:

$$(f - g)(2)$$

Answer: _____

7 Given $f(x)=x+4$, $g(x)=x^2-2$. Find:

$$(f \circ g)(4)$$

Answer: _____

8 Given $f(x)=x^2+1$, $g(x)=3x$. Find:

$$(f \circ g)(1)$$

Answer: _____

9 Given $f(x)=x-3$, $g(x)=x^2$. Find:

$$(f - g)(1)$$

Answer: _____

10 Given $f(x)=x+4$, $g(x)=x^2-2$. Find:

$$(f \circ g)(2)$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES $(f \circ g)(x)$ means $f(g(x))$ — substitute g first. Order matters: $f \circ g \neq g \circ f$ in general.

1 Given $f(x)=2x+3$, $g(x)=x-1$. Find:

$$= 11$$

$$(f + g)(3)$$

2 Given $f(x)=x^2+1$, $g(x)=3x$. Find:

$$= 37$$

$$(f \circ g)(2)$$

3 Given $f(x)=x+4$, $g(x)=x^2-2$. Find:

$$= 23$$

$$(g \circ f)(1)$$

4 Given $f(x)=2x$, $g(x)=x+5$. Find:

$$= 17$$

$$(f + g)(4)$$

5 Given $f(x)=x-3$, $g(x)=x^2$. Find:

$$= -5$$

$$(f - g)(2)$$

6 Given $f(x)=x-3$, $g(x)=x^2$. Find:

$$= -5$$

$$(f - g)(2)$$

7 Given $f(x)=x+4$, $g(x)=x^2-2$. Find:

$$= 18$$

$$(f \circ g)(4)$$

8 Given $f(x)=x^2+1$, $g(x)=3x$. Find:

$$= 10$$

$$(f \circ g)(1)$$

9 Given $f(x)=x-3$, $g(x)=x^2$. Find:

$$= -3$$

$$(f - g)(1)$$

10 Given $f(x)=x+4$, $g(x)=x^2-2$. Find:

$$= 6$$

$$(f \circ g)(2)$$



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